

# Ideas Register — Consolidated, v1 to v32

One file for every parked idea, with its current status

This file replaces `ideas_v1` through `ideas_v18`. Versions 1–11 were cumulative (each carried the previous forward) and 12–18 were separate additions, so the same idea often appeared in several files at different stages. Here each idea appears once, with where it came from, where it is destined, and where it stands now.

**How status was determined.** Not from memory of the discussions. Each status was checked by searching the current paper, `Pattern_Arithmetic_v15_28.tex` (v15\_28 differs from v15\_22 only in the atom-structure passages and the meta-vector subsection, so earlier checks stand), and — for items destined elsewhere — the `Assembly_v1.tex` and `Inference_v1.tex` copies on hand (which may be older than your working copies; if so, the “not filed” marks below should be rechecked against yours). Entries from v19 onward are appended here; v19 to v32 are the first fourteen.

**Status vocabulary.** **CLOSED** — resolved and merged into the paper. **CLOSED (redirected)** — determined not to be the paper’s question; destination named. **RESOLVED, NOT MERGED** — fully worked out but waiting on a prerequisite. **DECIDED** — a ruling recorded; nothing to merge. **PARTLY MERGED** — some of it is in the paper. **OPEN** — still unresolved or unfiled. **PARKED** — verified but with no current home.

## 1 The gaps of ideas v1, as they now stand

Honest summary first, since this is what prompted the consolidation:

| Gap                                                    | Status                                                                                                                                                                                    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gap 1</b> — Mix across differing directions         | <b>CLOSED</b> in v15_21: refused, on the <code>chladni</code> evidence, with Bundle and Collapse remaining available. One small residual (R26 below).                                     |
| <b>Gap 2</b> — Location as a sort                      | <b>CLOSED</b> in v15_25: location is a meta vector, not a sort, by the cap test and the seat precedent. The taxonomy it waited on is now Section 2.11, and the open paragraph is removed. |
| <b>Gap 3</b> — Reducing a path by merging close actors | <b>CLOSED for Paper 1</b> in v15_22 (narrowed to the algebraic guarantee); the technique question is <b>redirected to Inference</b> and is not yet filed there.                           |

The two remaining items from v1 are both Inference-destined and neither is filed: the candlestick summary (verified, mechanism works) and the “Inference Mandelbrot” naming flag.

**The single most useful next step** is now the scope decision (R27, R28): whether Axiom 7’s metric catalogue and the costumes stay in Paper 1, and whether the paper splits. It also decides which of R12, R25 and R41–R43 still matter.

## 2 Register

| ID                                                   | Item                                                                       | Origin        | Destination       | Status                                                                                                                           |
|------------------------------------------------------|----------------------------------------------------------------------------|---------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <i>Closed</i>                                        |                                                                            |               |                   |                                                                                                                                  |
| R1                                                   | Mix across differing directions (Gap 1)                                    | v1, v3–v4, v7 | Paper 1           | <b>CLOSED</b> — refused; v15.21                                                                                                  |
| R2                                                   | Terminology realignment: Bundle/Remove, Collapse/UnCollapse, Mix/UnMix     | v8–v10        | Paper 1           | <b>CLOSED</b> — merged; v15.20                                                                                                   |
| R3                                                   | Symbols $\uplus, \setminus, \oplus, \ominus, \boxplus, \boxminus$          | v9–v10        | Paper 1           | <b>CLOSED</b> — merged ( $\ominus$ superseded)                                                                                   |
| R4                                                   | Collapse non-idempotent; $N$ = list length; $A^{(N)}$ ; UnCollapse algebra | v10           | Paper 1           | <b>CLOSED</b> — merged                                                                                                           |
| R5                                                   | Distinct count $D$ against total $N$                                       | v13           | Paper 1           | <b>CLOSED</b> — merged; v15.20                                                                                                   |
| R6                                                   | Reducing a path (Gap 3): algebraic guarantee only                          | v1, v6, v18   | Paper 1           | <b>CLOSED (redirected)</b> — v15.22                                                                                              |
| R7                                                   | $\tau_{\text{weather}}$ conflation                                         | v4            | —                 | <b>CLOSED</b> — correction recorded                                                                                              |
| R8                                                   | Same-sort region merge vs cross-sort                                       | v6            | Assembly; Paper 1 | <b>CLOSED</b> — same-sort solved in Assembly; cross-sort is R1                                                                   |
| R9                                                   | Internal link, terminology, cohesion audit                                 | v12           | Paper 1           | <b>CLOSED</b> — fixes in v15.23, carried forward                                                                                 |
| R10                                                  | Stale status lines and symbols in v1–v10                                   | v11           | ideas             | <b>CLOSED</b> — superseded by this file                                                                                          |
| R11                                                  | Inertia in Assembly via a mass-like weighting                              | v13           | Paper 2           | <b>CLOSED</b> — tested; fails                                                                                                    |
| <i>Decided</i>                                       |                                                                            |               |                   |                                                                                                                                  |
| R12                                                  | Physics names for atom metrics                                             | v13           | Paper 1           | <b>DECIDED</b> — keep $\mathcal{E}$ ; $N/D$ earned; “density” stays retired; $S/N$ parked                                        |
| R13                                                  | “uncollapse” (v5) collides with UnCollapse                                 | v11           | ideas             | <b>DECIDED</b> — call it Mix-provenance link                                                                                     |
| <i>Resolved but not merged, and its prerequisite</i> |                                                                            |               |                   |                                                                                                                                  |
| R14                                                  | Gap 2: location and time as meta vectors; registers; age derived           | v1, v17       | Paper 1           | <b>CLOSED</b> — merged in v15.25                                                                                                 |
| R15                                                  | Meta-vector taxonomy; $V_{\text{derived}}$                                 | v2, v23       | Paper 1           | <b>CLOSED</b> — merged in v15.25 as Section 2.11                                                                                 |
| <i>Open — meta vectors</i>                           |                                                                            |               |                   |                                                                                                                                  |
| R16                                                  | Link vectors                                                               | v2            | Paper 2           | <b>PARTLY MERGED</b> — $V_{\text{link}}$ defined in Paper 1 (v15.25); structures built from links left to Assembly and Inference |
| R17                                                  | Meta-vector grouping (promote a shared one)                                | v2            | Inference         | <b>OPEN</b> — not filed                                                                                                          |
| R18                                                  | Identity handle / Mix-provenance link                                      | v5, v11       | Papers 1, 2, Inf. | <b>OPEN</b> — still “proposed, not adopted”; $V_{\text{link}}$ now names targets by identifier                                   |

| ID  | Item                                                                                                                                                                                                            | Origin   | Destination                     | Status                                                                                                                                                                              |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R19 | Should Mix/Collapse skip expired contributions?                                                                                                                                                                 | v17      | Paper 1; Inf.                   | <b>PARTLY MERGED</b> — validity rule merged (v15_25); filtering by UnCollapse not merged                                                                                            |
| R45 | Meta-vector design questions G1–G14 (v23, v24): label identity, the level rule, mutation and links, vocabulary, timestamp encoding, sentinels, granularity, comparison per kind, core multiplicity, guidance    | v23, v24 | Paper 1                         | <b>CLOSED</b> — merged in v15_25, with the closure-unit sentence, the identity wording and the granularity fixed in v15_26; G11 and G14’s merge default not merged                  |
| R46 | Scalar meta vectors as vectors in planes; a registry of recommended planes (non-normative)                                                                                                                      | v27      | Paper 1 (Future Work catalogue) | <b>PARTLY MERGED</b> — form (scalar on a designated axis, no other constraint) and the catalogue entry merged in v15_27 and corrected in v15_28; the catalogue itself not populated |
| R47 | Expiry as read-time exclusion: determinism, counts, hierarchy, links to expired atoms                                                                                                                           | v27      | Paper 1                         | <b>CLOSED</b> — merged in v15_27                                                                                                                                                    |
| R48 | Two-level meta vectors: a result’s own new timestamp beside its cores’                                                                                                                                          | v27      | Paper 1                         | <b>PARTLY MERGED</b> — timestamp, link and expiry merged (v15_26, v15_27); atom-level location not                                                                                  |
| R49 | “Never lose identity”: value, individual, handle; Mix and UnMix                                                                                                                                                 | v27      | Paper 1                         | <b>CLOSED</b> — merged in v15_27; Bundle reading to confirm                                                                                                                         |
| R50 | Meta-vector consistency obligations: labels of results of $T$ , offset, Remove and Assembly; fixed conventions (time standard, axis and unit per scalar kind); per-sort $V_{\text{derived}}$                    | v30      | Paper 1                         | <b>OPEN</b> — consistency only; meaning is left to use                                                                                                                              |
| R51 | The Mona Lisa as the test case for mixed-sort cores (R36): one core per canvas point; a bowl of single-sort atoms aligned by position labels today; standards per sort; a planar position kind in the catalogue | v31      | Paper 1                         | <b>OPEN</b> — answered in principle; adoption of R36 pending                                                                                                                        |
| R52 | Colour positioned by $V_{\text{derived}}$ : colour must be addressed by direction, not phase; locator is direction plus atom coherence; $ V $ measures size                                                     | v31      | Paper 1                         | <b>OPEN</b> — convention to settle                                                                                                                                                  |

| ID                                         | Item                                                                                                                                                                                                                       | Origin   | Destination | Status                                                                                                                                   |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------|
| R53                                        | Review of v15_28, reproducibility exceptions: $\sigma_{\text{atom}}$ populations not printed; the 804-of-3000 count is rounding-sensitive; shell binning at floating-point edges; “checked directly” claims without inputs | v32      | Paper 1     | <b>OPEN</b> — supplement delivered; text fixes pending                                                                                   |
| R54                                        | Review of v15_28, mechanical consistency fixes: stale “not a sixth verb”; Table 1 says “contribution” for UnCollapse; title banner overflows; a drafting-history sentence in Axiom 7; “mass” wording in the enclosure      | v32      | Paper 1     | <b>OPEN</b> — v15_29                                                                                                                     |
| R55                                        | The bowl as a holder of $V_{\text{derived}}$ , the metrics and meta vectors: one statement over a holder                                                                                                                   | v32      | Paper 1     | <b>OPEN</b> — atoms only today                                                                                                           |
| <i>Open — Collapse</i>                     |                                                                                                                                                                                                                            |          |             |                                                                                                                                          |
| R20                                        | Collapse across differing sorts (per-vector $\tau_i$ )                                                                                                                                                                     | v10, v11 | Paper 1     | <b>OPEN</b> — stated in paper; broadened by R36                                                                                          |
| R21                                        | Collapse as $T$ ’s actor                                                                                                                                                                                                   | v11      | Paper 1     | <b>OPEN</b> — “not claimed” in Axiom 7; see R40                                                                                          |
| R22                                        | Assembly and Collapse: relation                                                                                                                                                                                            | v11      | Papers 1, 2 | <b>CLOSED</b> — Axiom 7 already says $V_{\text{derived}}$ stands in for the atom in Assembly; Axiom 6 does not repeat it (corrected v26) |
| <i>Open — metrics, scope, presentation</i> |                                                                                                                                                                                                                            |          |             |                                                                                                                                          |
| R23                                        | Shell-indexed energy                                                                                                                                                                                                       | v14      | Paper 1     | <b>OPEN</b> — undefined                                                                                                                  |
| R24                                        | Antipodal ( $\hat{n}$ against $-\hat{n}$ ) cancellation                                                                                                                                                                    | v14      | Paper 1     | <b>OPEN</b> — special case of closed R1                                                                                                  |
| R25                                        | Separable-metric pointwise test                                                                                                                                                                                            | v15      | Paper 1     | <b>PARKED</b> — verified; value depends on R27                                                                                           |
| R26                                        | Route-1 scoping clause (refusal is about $\boxplus$ only)                                                                                                                                                                  | v7       | Paper 1     | <b>PARTLY MERGED</b> — one sentence not carried; closure wording “already demonstrated ... correct behavior” overstates (v22)            |
| R27                                        | Axiom 7 metric catalogue and costumes: cut or move?                                                                                                                                                                        | v16      | Paper 1     | <b>OPEN</b> — undecided                                                                                                                  |
| R28                                        | Two-paper split                                                                                                                                                                                                            | v16      | Papers 1, 2 | <b>OPEN</b> — undecided                                                                                                                  |
| R29                                        | Review’s presentational items                                                                                                                                                                                              | v16      | Papers 1, 2 | <b>OPEN</b> — not acted on                                                                                                               |
| <i>Open — Inference-bound</i>              |                                                                                                                                                                                                                            |          |             |                                                                                                                                          |
| R30                                        | Candlestick summary                                                                                                                                                                                                        | v1       | Inference   | <b>OPEN</b> — verified; not filed                                                                                                        |
| R31                                        | “Inference Mandelbrot” naming flag                                                                                                                                                                                         | v1       | Inference   | <b>OPEN</b> — flag only                                                                                                                  |
| R32                                        | Technique selection for reduction (average, centroids, shape-preserving)                                                                                                                                                   | v18      | Inference   | <b>OPEN</b> — specified; not filed                                                                                                       |
| R33                                        | Douglas–Peucker adapted to the sphere                                                                                                                                                                                      | v18      | Inference   | <b>OPEN</b> — the one unbuilt piece                                                                                                      |

| ID                                              | Item                                                                                                                                                                                  | Origin | Destination | Status                                                                                                             |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|--------------------------------------------------------------------------------------------------------------------|
| R34                                             | Sphere-packing bound for the “stated distance”                                                                                                                                        | v6     | Inference   | <b>OPEN</b> — not filed anywhere                                                                                   |
| R35                                             | Coherence check for the candlestick                                                                                                                                                   | v1     | Inference   | <b>OPEN</b> — never considered                                                                                     |
| <i>Open — atom structure (v19)</i>              |                                                                                                                                                                                       |        |             |                                                                                                                    |
| R36                                             | Pure and mixed atoms: cores; per-vector $\tau_i$ as the general form; Mona Lisa as test case; direction-based derived quantities per sort                                             | v19    | Paper 1     | <b>OPEN</b> — broadens R20                                                                                         |
| R37                                             | Collapse of multi-vector contributions (not fixed at one vector each)                                                                                                                 | v19    | Paper 1     | <b>CLOSED</b> — cores of any size, kept distinct under Collapse; merged in v15_24                                  |
| R38                                             | $N$ as number of cores; $Q$ as number of vectors (resolves the collision with $N$ as vector count in Axiom 7)                                                                         | v19    | Paper 1     | <b>CLOSED</b> — $N$ cores, $Q$ vectors; merged in v15_23                                                           |
| R39                                             | Collapsed atom as a counted bowl; commuting test                                                                                                                                      | v19    | Paper 1     | <b>OPEN</b> — test not yet run                                                                                     |
| R40                                             | $T$ asymmetry: a bowl is not an actor; a Collapsed atom needs a cast                                                                                                                  | v19    | Paper 1     | <b>OPEN</b> — relates to R21                                                                                       |
| R44                                             | Empty atom: black is $N = 1$ , $Q = 0$ ; empty core under Collapse; zero-strength vector is not an arrow; conventions at $Q < 2$                                                      | v22    | Paper 1     | <b>CLOSED</b> — merged in v15_24; “info number” label moved to $Q$ (confirm)                                       |
| <i>Open — energy and effective counts (v20)</i> |                                                                                                                                                                                       |        |             |                                                                                                                    |
| R41                                             | Energy from strength, vector count and direction:<br>$\mathcal{E} = S^2/Q + Q \text{Var}(r)$ ;<br>$S^2/Q \leq \mathcal{E} \leq S$ ;<br>$ V ^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$ | v20    | Paper 1     | <b>PARKED</b> — identities verified, not in paper; $\mathcal{E}$ keeps “energy,” $\mathcal{V}$ keeps “volume”      |
| R42                                             | Effective vector count<br>$Q_{\text{eff}} = S^2/\mathcal{E}$ ; overlap with $\sigma_{\text{atom}}$                                                                                    | v20    | Paper 1     | <b>CLOSED</b> — merged in v15_23 as a re-expression, not an eighth metric; both it and $\sigma_{\text{atom}}$ kept |
| R43                                             | Strength-only metrics per sort and for the whole; effective number of sorts $1/\sum w_s^2$ ; pooling ratio                                                                            | v20    | Paper 1     | <b>OPEN</b> — needs one stated convention                                                                          |

### 3 The entries, by version

#### v1 — the three gaps and two Inference items

**Gap 1** (Mix across differing directions): Axiom 5 defines Mix only at a shared  $\hat{n}$ ; a provisional vector-sum recipe had been tried and withdrawn. *Status: R1, closed.* **Gap 2** (location as a sort): a third candidate was tested — disjoint dimensions for content, time, and location. Verified: two shifts through disjoint dimensions commute exactly (to full floating-point precision), and content stays recoverable by projection; shifts forced onto one shared dimension do not commute. Left

open: the encoding itself. *Status: R14, resolved in v17.* **Gap 3** (reducing a path): needed a distance, a merge rule, and a drift bound. Verified that swapping the clip for the Hill form  $r' = |z|/(K+|z|)$  changes the strength but not  $\chi' = \arg(z)$ . *Status: R6, closed (redirected).* **Candlestick summary:** open/close/high/low distinguishes two sequences of identical content in opposite order that naive Mix cannot — Mix gave the same  $47.5^\circ$  for both; the candle’s open and close swapped ( $20^\circ \rightarrow 160^\circ$  against  $160^\circ \rightarrow 20^\circ$ ) while high and low stayed equal. *Status: R30, 0 occurrences in Inference\_v1.* **“Inference Mandelbrot”:** an ordered-thing promotion is a different problem from Assembly’s Hill-bijection promotion of unordered camps, and needs its own name. *Status: R31.* The “considered and found to need nothing filed” list (reference atoms, precursor-witness relabeling, the  $\pi = (x; w_1, \dots, w_n)$  trajectory,  $\chi$  as internal phase-color) was rechecked in v11 against the current paper: all present, nothing to file. *Closed.*

## v2 — meta vectors, split across three papers

Proposed: besides information vectors (capped,  $r_i \leq 1$ ), an atom may carry meta vectors describing it rather than being part of its content:  $V_{\text{derived}}$ , link vectors, a timestamp vector. Meta vectors are not capped. Split by destination: the taxonomy and  $V_{\text{derived}}$  to Paper 1; link vectors to Assembly; the grouping mechanism (query nearby meta vectors, promote one shared value) to Inference. Verified caveat: grouping  $V_{\text{derived}}$  inherits the Hill form’s limit — both fix magnitude only, never direction. *Status: R15–R17. The paper has zero occurrences of “meta vector”; Assembly\_v1 has zero of “link vector.”*

## v3 — Gap 1, characterized

Argued reading: Mix takes several reports already agreed to answer one shared question and returns a verdict about it; the shared  $\hat{n}$  is what makes that true. Notably, v3 already stated that in Mix’s domain “ $\hat{n}$  itself was never plural; only  $\chi$  ever varied” — the very distinction that had to be rediscovered during the Gap 3 discussion (v18) before it stopped causing confusion. Four “we mix all the time” counterexamples were checked and dissolved (sugar in tea is one dial; red plus green is already same- $\hat{n}$ ; paint compositing is a bowl and flattening is destructive and irreversible — matching UnMix’s impossibility). The one unresolved example, a whole painting as one vector, is Gap 1 itself. The open question split in two: *embedding validity* (does closeness on a shared sphere track semantic agreement?) and the *merge rule*. *Status: superseded by R1’s closure — Mix refuses, so neither sub-question needs an answer for Mix.*

## v4 — the $\tau_{\text{weather}}$ correction; Gap 1 “ready to close”

$\tau_{\text{weather}}$  had named two different problems: combining different weather concepts across directions (Gap 1), and reducing closely aligned same-sort atoms to one representative, which is Assembly’s already-verified Hill-bijection mechanism (16 camps to 13 systems). The **chladni** section was found to already contain the evidence for Gap 1 (“cross-mode coexistence is therefore  $\oplus$ ”). *Status: R7 closed; the chladni evidence is what the Gap 1 closure rests on.*

## v5 — identity handle

An opaque *id* meta field, assigned before Bundle or Mix and kept after, so a result can point back to the bowl that produced it (bowl {red, green} is #17; Mix gives yellow #18 with a link  $\#18 \rightarrow \#17$ ). Refuse  $\tau_{\text{id}}$  as a sort; keep *id* as a field. Navigation by a kept record, not inversion of Mix. Renamed in v11 “Mix-provenance link” because the original word “uncollapse” collides with

the formal UnCollapse operator. *Status: R13 decided; R18 open — the paper mentions  $id_i$  once, as “proposed, not adopted,” in UnCollapse’s well-definedness paragraph.*

## **v6 — sphere-packing bound; merging a whole region**

For Gap 3’s “stated distance”: the number of mutually distinguishable points at separation  $\theta_{\min}$  is roughly  $2/(1 - \cos(\theta_{\min}/2))$  — about 52,525 at  $1^\circ$ , 2,101 at  $5^\circ$ , 526 at  $10^\circ$ , 59 at  $30^\circ$ , 15 at  $60^\circ$ , 7 at  $90^\circ$ , 4 at  $120^\circ$ . Merging a whole region to one representative splits: same-sort is solved (Hill bijection) and cross-sort is Gap 1. *Status: R8 closed; the bound itself is R34, filed nowhere.*

## **v7 — scoping of Route 1**

“Refuse outright” should be worded as a claim about  $\boxplus$ , not a permanent bar on any future, separately admitted mechanism — the same “undecided, not admitted” logic Axiom 0 already uses. *Status: R26, partly merged.* Checked against the Gap 1 closure: the closure is scoped to Mix and keeps  $\boxplus$  and  $\oplus$  available, which honours the point. What was not carried is the one explicit sentence that a separately admitted mechanism is not barred, and the closure calls the other two routes “withdrawn,” slightly stronger than v7 intended. An optional one-sentence fix.

## **v8, v9, v10 — terminology and verb inventory**

v8 proposed the Collapse/UnCollapse/Bundle/Mix/UnMix realignment and flagged a hard symbol collision ( $\ominus$  was Remove). v9 chose  $\boxplus$  for Bundle and  $\ominus$  for UnCollapse, and scoped ids to the one case that needs them: duplicate-valued contributions in UnCollapse. v10 revised the symbols —  $\ominus$  to UnCollapse, Remove to  $\backslash$  — settled Bundle’s idempotence ( $A \boxplus A = A$ , already proven), and showed Collapse is *not* idempotent by three convergent arguments (information loss; an algebraic contradiction with  $A \text{ UnCollapse } A = 0$ ; the same evidence-count blindness the Hill form was built to fix).  $N$  is simply the length of the vector list;  $A^{(N)}$  abbreviates  $N$  identical copies; UnCollapse sends  $N \rightarrow N - 1$ . *Status: R2–R4 closed, all merged. MixMerge was later reverted to Mix at your request;  $\ominus$  appears zero times in the paper. The per-vector  $\tau_i$  question that v10 left open is R20.*

## **v11 — review of v10 against the paper**

Status lines of v8–v10 corrected (the realignment was merged); stale  $\oplus/\ominus$  readings in earlier text fixed, including one sentence that had become *false* (“Gather still satisfies  $A \oplus A = A$ ” reads wrongly once  $\oplus$  means Collapse); the “uncollapse” collision resolved by renaming. Three gaps opened by giving Collapse axiom-level standing: R20 (across sorts), R21 (as an actor for  $T$ ), R22 (no stated relation to Assembly — Axiom 6 mentions Collapse zero times, so this one is silence rather than a stated omission). One new route recorded: Collapse losslessly first, then define a reading — the seed of what v18 developed. *Status: R10, R13 closed or decided; R20–R22 open.*

## **v12 — internal audit**

96 labels and 198 reference call-sites checked, each against what its sentence claims; zero mismatches. Two stragglers fixed (an Abstract “a Collapse” noun phrase; a Forest-costume “gathers”). Four cohesion fixes where the paper’s own summaries omitted Collapse, and the UnMix proof section now names UnMix. No new conceptual gap. *Status: R9 closed; fixes are in the current paper.*

### v13 — naming the atom metrics

Seven metrics, quoted precisely. Earlier drafts had called  $\mathcal{V}_{\text{atom}}$  “mass,”  $\gamma_{\text{atom}}$  “density,” and  $S_{\text{atom}}$  “atom energy,” and retired all three; only  $\mathcal{E}_{\text{atom}} = \sum r_i^2$  kept “energy,” because it is tied by an exact identity to  $|V_{\text{derived}}|^2 = \mathcal{E}_{\text{atom}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)$  (variance-of-a-sum structure, verified to six decimals). The rule: a physics name is kept only if the quantity participates in something provable.  $D = |A_1 \uplus \dots \uplus A_n|$  passes (forced by  $\uplus$ ’s idempotence), so  $N$  and  $D$  keep the mass-number/atomic-number analogy; “density” stays retired;  $S_{\text{atom}}/N$  is parked rather than named. Inertia in Assembly is closed: a mass-like weighting was already tested in the motion equation — a ninefold strength gap gives a 729-fold weight and a non-monotonic trajectory. *Added in conversation after v15, not previously filed:* asked whether either of the two metrics survives, the answer was that energy already does ( $\mathcal{E}_{\text{atom}}$ , earned) while no candidate for density does —  $S_{\text{atom}}$  already has a name,  $N$  is a count rather than a ratio, and the one ratio with the right shape,  $\gamma_{\text{atom}}$ , is the one already rejected. *Status: R5 closed; R11 closed; R12 decided.*

### v14 — three rough sketches

*Density* restated: nothing new, closed by reference to v13. *Energy as shell distribution:* partly already true of  $\mathcal{E}_{\text{atom}}$  — for fixed total strength  $S = 2.0$ , four vectors give  $\mathcal{E} = 1.0$  spread evenly and 1.81 concentrated toward the extremes (and  $\sigma_{\text{atom}}$  goes  $0 \rightarrow 0.333$ ) — but  $\sigma_{\text{atom}}$  is blind to *which side* the concentration is on, and no shell-indexed quantity exists. *Matter/antimatter:* antipodal cancellation ( $\hat{n}$  against  $-\hat{n}$ ), a second kind of opposition beside the existing same-direction opposite-phase cancellation. With Gap 1 now closed by refusal, this would be a deliberate special-case exception, parallel to the existing one; what the rule would produce, and whether it needs equal strength, is undecided. *Status: R23, R24 open.*

### v15 — when two metrics constrain each other

For separable metrics  $M_1 = \sum f(r_i)$  and  $M_2 = \sum g(r_i)$ : if  $f \leq g$  everywhere on  $[0, 1]$  then  $M_1 \leq M_2$  for every atom; if the curves cross there is no bound. Verified against both cases (it forces  $\mathcal{E} \leq S$  because  $r^2 \leq r$ ; it forces nothing for  $N$  against a shell-weighted energy). Where a bound holds, the gap is itself a metric ( $r - r^2$ , largest at  $r = 0.5$ ). Possible uses (a non-redundant reporting set, a two-axis placement, a debugging check, a post-hoc label for Assembly’s camps) are speculative. *Status: R25, parked; it matters mainly if R27 keeps the metrics.*

### v16 — external review and the two-paper split

Every checkable claim in the review was verified: the worked numbers (e.g.  $r' \approx 0.721$  at  $73.9^\circ$ ; freeze at 628 and 1261 steps), that all 22 occurrences of the six Axiom 7 metrics sit inside Axiom 7 and none in Bundle/Mix/T/offset, that “white” is used in two senses (the paper already says so in Axiom 3), and that the “checked directly” precision claims have no supplement. The proposal: Paper 1 is the typed algebra, Paper 2 is assembly and the enclosure, the Axiom 7 catalogue and the costumes go in neither; Paper 1 states Assembly as a signature only and ends by naming the seat it refuses to supply; cut by question, not by length. *Not itemized when v16 was filed, but in the review and still pending (R29):* state the clip, the Mix refusal, and the assembly energy as stipulated choices in one place rather than narrated as discoveries; print the enclosure gradient; provide a supplement for the “checked directly” claims; and stop calling the enclosure a “law” in the same voice as  $\text{SO}(3)$  while it still needs  $\delta$ ,  $\eta$ , the census cut, and  $\varepsilon_{\text{allow}}$ . *Status: R27, R28, R29 open and undecided.*

## v17 — Gap 2 resolved

Meta vector, not sort, tested against the cap:  $r \leq 1$  encodes strength of content, while location and time are coordinates. This is the line the paper already draws for Assembly’s seat. Location is one vector via standard  $(\theta, \phi)$  from latitude and longitude (London  $\rightarrow (0.6225, -0.0011, 0.7826)$ , Tokyo  $\rightarrow (-0.6194, 0.5252, 0.5835)$ , Sydney  $\rightarrow (-0.7273, 0.3999, -0.5577)$ ). Time on  $(\theta, \phi)$  fails: at hour 0 every minute lands on  $(0, 0, 1)$ , and 11:59 and 0:00 land about 2 apart (antipodal). The fix: minute on  $\phi$ , hour on  $\chi$ ,  $\theta$  fixed at the equator ( $6^\circ$  and under  $1^\circ$  apart for the same pair). Timestamp is a register of length  $N$ , kept in step with UnCollapse; age is derived (now – timestamp), never stored; expiry is a second register because it cannot be derived; timestamps help but do not guarantee disambiguation the way an *id* would. **Blocking:** cannot be merged until R15 is. *Status: R14, R19.*

## v18 — Gap 3 closed for Paper 1

Corrected an earlier confusion ( $\chi$  is phase at a shared direction, not  $\hat{n}$ ; Gap 1’s refusal did not apply). The real issue is order-blindness: merging three actors in either order gives the same result. A path’s commitment to order comes from outside the algebra — Bundle discards order, and choosing  $T$  over a bowl is the declaration that order matters. So averaging 10,000 temperature readings is bowl-plus-Mix and loses nothing. Three techniques answer three different questions: averaging cannot tell a spike from a flat line; centroids give identical results for a sequence and its reverse (clustering is blind to position); shape-preserving reduction keeps real points at their positions — Douglas–Peucker, whose tolerance is the drift bound Gap 3 asked for. Paper 1 was narrowed to the non-consumption guarantee. *Status: R6 closed (redirected); R32, R33 open for Inference.*

## v19 — pure and mixed atoms, cores, and Collapse against a bowl

Prompted by a hypothetical: could the Mona Lisa be held in one atom? “One atom” can mean three things. A *name*: a single seated point, like “storm,” which the paper calls a doorbell — it carries none of the content. A *bowl*: already legal for mixed sorts, since Bundle is sort-blind. A *Collapsed atom* holding colors, words, and strokes, which needs mixed sorts inside one atom. This is the example v3 left unresolved (“a whole painting as a single vector” was Gap 1 restated). With Gap 1 closed, a single *vector* is refused, while one atom of many vectors stays available. What stands in the way of the third meaning: mixed sorts (R20, widened to R36); where each contribution sits on the canvas — a location per contribution, the same kind of register as v17’s but not yet specified per contribution; derived quantities across sorts; and relations, since the paper “does not mint the relations” and, by its own design rule, which units were co-present and within what region are facts of a level, not coordinates of an atom.

## Rulings given in the discussion, none yet in the paper.

- *Collapse juxtaposes.* Positions do not change, nothing merges, no new type is minted. Relations within a sort are as before; across sorts they exist only under a stated rule (Axiom 0’s “undecided, not admitted,” applied to Collapse). So derived quantities that use direction are computed per sort, and cross-sort ones are undefined until a rule exists. (Refined in v20: strength-only quantities may also be given for the whole atom.)
- *Cores.* A core is a unit inside an atom, usually one  $\tau$  (a pure type) but free to hold several — words and numbers together — as typical, not as a rule. The general form is per-vector  $\tau_i$ , with “core” a grouping. “Core” appears in the paper only in the “Core Contributions” heading and in “no core, no halo” in the enclosure, so it needs one defined sense.
- *Collapse is not fixed at one vector per contribution.* This settles the question the paper was silent on: whether collapsing already-multi-vector atoms flattens or nests. The Collapse text still says “one vector per contribution.”
- *$N$  is the number of cores.* This keeps UnCollapse’s  $N \rightarrow N - 1$ , keeps  $D$  a bowl size (distinct cores), and keeps the v17 registers at length  $N$ , so those registers are per core, not per vector. The cost is a word collision:  $N$  is currently the vector count in Axiom 7, in  $\gamma_{\text{atom}} = \mathcal{V}_{\text{atom}}/N$  and in  $\sigma_{\text{atom}}$ ’s  $N^2/(N - 1)$ . *Decided:*  $N$  counts cores and  $Q$  counts vectors. All seven Axiom 7 metrics run over vectors, so they take  $Q$ ; the  $D/N$  paragraph and the UnCollapse text keep  $N$  and change “vectors” to “cores.” To settle at the merge: the sentence that defines  $N$  as the atom’s count of informational vectors, and where the label “info number” goes.
- *A Collapsed atom matches a counted bowl.* They are identical exactly when nothing repeats ( $D = N$ ), and  $N - D$  is the gap. The paper’s bowl is a set, and its own counting equation shows the difference:  $A \uplus A \uplus A \uplus A \setminus A = \emptyset$ , while  $A \oplus A \oplus A \oplus A \ominus A = A^{(3)}$  (by the UnCollapse rule). Operations should work on both; recorded as a commuting test — a pack/unpack map between the two, with every operation required to agree across it. Sum-style quantities ( $S$ ,  $\mathcal{E}$ ,  $V_{\text{derived}}$ ) agree only on a counted bowl.  $T$  is asymmetric today: it “does not act on the set  $R$  as a set,” and a Collapsed atom can act only once a cast is named.
- *Two distinct ways of arranging and viewing vectors.* They differ in multiplicity (set against counted), in handle (a Collapsed atom is one unit that can sit in a bowl or at a lexicon pin; a bowl is a collection), and in sequence (neither stores order; the path carries it, and a bowl is a good surface for displaying one).

*Status: R36–R40; R20 and R21 cross-referenced.*

## v20 — energy from strength, counts, and direction; effective counts

Started from a recollection that energy is  $\sum r^3$ . The paper’s energy is  $\mathcal{E} = \sum r^2$ ;  $\sum r^3$  is the atom’s volume, formerly “mass.” Nothing needs renaming, and  $\sum r^3$  keeps a geometric reading (strength as a radius, as in Assembly’s softening length).

**Energy from strength, vector count  $Q$ , and direction already exists in  $\mathcal{E}$ .** Three exact relations, verified on 20,000 random atoms (errors near  $10^{-14}$ , no violations), with  $S = \sum r_i$ :

- $\mathcal{E} = S^2/Q + Q \text{Var}(r)$ : an evenly spread floor, set by  $S$  and  $Q$  alone, plus concentration.
- $S^2/Q \leq \mathcal{E} \leq S$ .

- $|V_{\text{derived}}|^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$ : the resultant’s energy, blending the incoherent baseline with the fully aligned value, weighted by the paper’s coherence  $\alpha$ .

Which works better: the floor  $S^2/Q$  cannot tell an even atom from a concentrated one (both give 1.000 at  $S = 2$ ,  $Q = 4$ , where  $\mathcal{E}$  gives 1.000 against 1.810).  $\mathcal{V} = \sum r^3$  separates them even more sharply (0.500 against 1.715) but has no identity tying it to anything beyond the radius reading. So  $\mathcal{E}$  is the energy and  $\mathcal{V}$  stays volume. Cores  $N$  enter through the same variance-of-a-sum algebra applied per core (plain algebra, not run separately) and through the ratios  $Q/N$  and  $S/N$ . The mean strength  $S/Q$ , the parked candidate from v13 (then written  $S/N$ ), is the centre of the variance decomposition and keeps a plain name.

**Effective vector count**  $Q_{\text{eff}} = S^2/\mathcal{E}$  (the participation ratio in physics, the effective sample size in statistics).

- $Q$  counts vectors and  $Q_{\text{eff}}$  counts those that matter: 100 vectors at 0.05 give 100; one vector at 0.9 plus 99 at 0.001 give 1.23.
- The largest vector carries at least  $1/Q_{\text{eff}}$  of the strength (50,000 random atoms, no violations).
- Scale-free: halving every  $r$  leaves it unchanged (2.2099 both).
- With direction:  $|V|^2/S^2 = \alpha + (1 - \alpha)/Q_{\text{eff}}$  (error near  $7 \times 10^{-16}$ ), so two dimensionless numbers fix the resultant’s size.
- Limits: it re-expresses  $S$  and  $\mathcal{E}$ , adding no information, and ignores direction.
- Overlap:  $\sigma_{\text{atom}} = (Q/Q_{\text{eff},\mathcal{V}} - 1)/(Q - 1)$ , where  $Q_{\text{eff},\mathcal{V}}$  is the effective count of the *cubed* shares (error near  $10^{-15}$ ). So  $\sigma_{\text{atom}}$  is already this family, and  $Q_{\text{eff}}$  is its plain-strength sibling, the one tied by identity to  $\mathcal{E}$  and  $|V|^2$ . Whether to keep both, one, or neither is undecided.

**Per sort and for the whole.** The per-sort rule belongs to quantities that use direction ( $V_{\text{derived}}$ ,  $\alpha$ ,  $|V|^2$ ), since sums and dot products of directions across sorts mean nothing. Strength-only quantities ( $S$ ,  $\mathcal{E}$ ,  $\mathcal{V}$ ,  $Q_{\text{eff}}$ , the  $\sigma$ ’s) can be given per sort and for the whole, on one stated convention: each sort’s  $r$  is normalized to its own declared saturation (the paper’s  $r = a/a_{\text{max}}$ ), and whole-atom figures treat fraction of own saturation as a common scale. With  $w_s$  the share of total strength held by sort  $s$ :

- Pooling ratio  $\sum_s Q_{\text{eff},s}/Q_{\text{eff,whole}} \geq 1$  (Cauchy–Schwarz; no violations in 30,000 atoms), with equality exactly when every sort has the same strength-weighted mean strength  $\mathcal{E}_s/S_s$ . It measures imbalance between sorts: two vectors at 0.9 against ten at 0.1 give 2.63.
- Exact decomposition:  $1/Q_{\text{eff,whole}} = \sum_s w_s^2/Q_{\text{eff},s}$  (error near  $2 \times 10^{-16}$ ). In the symmetric case it multiplies: three sorts of four vectors at 0.5 give  $12 = 3 \times 4$ .
- Effective number of sorts  $1/\sum_s w_s^2$  answers “is this atom really mixed?” more readably than the ratio: for 50 color vectors at 0.5, 3 word vectors at 0.5, and 10 stroke vectors at 0.3,  $Q_{\text{eff}}$  is 50, 3, 10 per sort, 61.5 for the whole, and the effective number of sorts is 1.37. Three sorts are present but the atom reads as about 1.4, because color carries 85% of the strength.

*Status: R41–R43; the v19 per-sort line is refined above.*

## v21 — merging $N$ , $Q$ , and $Q_{\text{eff}}$ into the paper

Merged into `Pattern_Arithmetic_v15_23.tex`. Before any edit, all 62 standalone uses of  $N$  were audited and classified, since a blind replacement would also have hit  $N_{\text{dim}}$  and  $N_{\text{vox}}$ .

**What went in.** “Core” is defined once, at Axiom 7’s opening:  $N$  counts cores (the atom’s info number) and  $Q$  counts informational vectors, with  $Q \geq N$ . Every formula that runs over vectors takes  $Q$ : the  $V_{\text{derived}}$  sum,  $S$ ,  $\mathcal{V}$ ,  $\gamma$ ,  $\mathcal{E}$ , the coherence and concentration discussions,  $\sigma_{\text{atom}}$ ,  $\sigma_{\text{shell}}$ , the higher-dimensional actor condition  $Q \geq N_{\text{dim}} - 2$ , and the closing teaching-object sentence. The Collapse, UnCollapse and  $D/N$  text keeps  $N$  and now says “cores,” and UnCollapse states that  $Q$  falls by the removed core’s vector count.  $Q_{\text{eff}}$  is defined at the end of Axiom 7 as a re-expression of  $S$  and  $\mathcal{E}$ , with its range, its scale-freeness, the largest-vector bound, the count-versus-evenness distinction, and its exact relation to  $\sigma_{\text{atom}}$ .

**Found along the way.** The audit turned up more than the rename:

- $N$  was also the clock modulus in the numeric-sort definition, while the later clock section already used  $M = 24$ . The general form now uses  $M$ .
- “Core” was already in the paper twice: the heading “Core Contributions” (now “Principal Contributions”) and “no core, no halo” in the enclosure (now “no dense center, no halo”).
- A stray noun use, “the actor’s Collapse,” survived every earlier terminology sweep because the search pattern missed possessives. It now reads “Collapsed atom.”
- $\sigma_{\text{atom}}$  is exactly the normalized Herfindahl index of the cubed shares,  $(QH - 1)/(Q - 1)$ , and the paper had already rejected the unnormalized index for reading an even atom as  $1/Q$ . This is why  $Q_{\text{eff}}$  is described as a *count*, with  $Q_{\text{eff}}/Q$  as the size-free form. It sharpens v20, which called the two overlapping: they are the same construction on different shares.

**Deliberately not merged.** Mixed sorts, per-vector  $\tau_i$  (R36; R20 stays open and the Collapse section still requires a shared sort); collapsing atoms that already hold several cores, now stated in the paper as not defined; the per-core registers (R14, R15); and the other R41 identities.

**Checks.** The 18 numbered equations before Process Principles are identical to v15\_22’s after mapping  $Q$  back to  $N$ , so the math changed only by the symbol. Zero errors, zero undefined references, and the same five overfull warnings as before. One choice to confirm: the label “info number” was attached to  $N$ . *Status: R38 and R42 closed; R37 partly merged.*

## v22 — cores of any size, the empty atom, and a review of v15\_23

Merged into `Pattern_Arithmetic_v15_24.tex`. The trigger was an external critique of v15\_23, checked claim by claim against the text before anything changed. (The v21 entry above reached this file only now; it describes v15\_23, which this entry supersedes where they differ.)

**The critique, as it came out.**

- *Confirmed.* No example separated  $N$  from  $Q$  and, stronger than the critique said, no construction in the paper could produce  $Q > N$ : the lexicon cast seats single points, and collapsing multi-vector atoms was “not defined here.” The UnCollapse well-definedness paragraph and Table 2’s footnote still spoke of vectors where cores were meant.  $D$  was undefined for multi-vector cores: with  $A = \{0.8, 0.4\}$  and  $B = \{0.6\}$ ,  $\text{Collapse}(A, B, A)$  has  $D = 2$  read as a bowl and  $D = 3$  read as distinct vectors.
- *Half right.* The nucleon analogy: nucleon = core was the intended mapping, but the sentence did not say so.

- *Not valid.* That clipping is narrated as a finding: the paper has a heading “How the bound is imposed is a choice.”
- *Valid, and my wording.* The Mix-refusal closure calls refusal “already demonstrated, in a concrete domain, to be the correct behavior,” which overstates what `chladni` shows. Recorded against R26.
- *Missed by the critique, found in the check.* The five-fold cone sentence was ambiguous: “the same half-angle” reproduces the printed 7.14 and 0.988 only at half-angle  $5.7^\circ$ , while the pair-centred reading gives 7.203 and 0.997. Also,  $n$  and  $N$  both named the operand count, and  $Q_{\text{eff}}$  was undefined at  $S = 0$ .

**Process note.** The v21 audit searched for  $N$  and not for the word “vector,” so passages that never contain  $N$  were invisible to it. This round searched for both.

**Rulings applied.**  $Q \geq N$  is the usual case but not a rule, and  $N = Q = 1$  is the teaching atom, one state among several. An atom may hold no vector at all: the empty atom is black, one seat and no arrow,  $N = 1$ ,  $Q = 0$ , which is Axiom 2’s own wording. Three consequences were confirmed: black is a counted contribution under Collapse ( $A \oplus 0$  adds a core and no vectors, so  $0 \oplus 0 \neq 0$ ); Collapse keeps cores distinct ( $N = \sum_i N_i$ ,  $Q = \sum_i Q_i$ ), which answers flatten against nest; and UnCollapse at  $N = 1$  leaves black, the one exception to  $N \rightarrow N - 1$ .

**What changed in the paper.** Axiom 7’s opening restated around cores; a sentence in Axiom 2; the Collapse definition; non-idempotence stated for every  $A$ , black included, with  $A^{(k)}$  replacing  $A^{(N)}$ ; a paragraph on silent contributions;  $D$  defined by core identity (two cores are one value exactly when  $\uplus$  would seat them together, and empty cores coincide); a handle per core; the analogy wording; Table 2’s footnote; two summary sentences; the cone’s half-angle;  $Q_{\text{eff}}$  stated for  $S > 0$ ; an “empty atom” paragraph of conventions; and a worked count,  $N = 3$ ,  $Q = 5$ ,  $D = 2$ , with UnCollapse leaving  $N = 2$ ,  $Q = 3$  and a silent contribution giving  $N = 4$ ,  $D = 3$ ,  $Q = 5$ . The count is given in counts only: a first draft quoted strengths and put the seven metrics outside Axiom 7, undoing the isolation the review verified, so it was reverted.

**Two consequences applied that were not discussed, to confirm.** First, the label “info number” now goes with  $Q$ , since  $Q$  counts the information and adding an empty core does not change it; v21 had attached it to  $N$ . Second, informational vectors are positive,  $0 < r_i \leq 1$ : a vector of strength zero is black and not an arrow, so it is the empty core and does not count in  $Q$ . This follows from Axiom 2 and avoids a zero-strength vector counting in  $Q$  while black does not.

**Checks.** All 49 numbered equations are identical to v15\_23’s; zero errors and undefined references; the same five overfull warnings as before; the seven metrics and  $Q_{\text{eff}}$  occur zero times outside Axiom 7. *Status: R37 closed; R44 closed; R26 extended.*

## v23 — meta vectors as specified, and what to settle before importing them

Specification as given, parked here and not yet in the paper:

- Meta vectors are labels on coaches; information vectors are the content in the coaches. Meta vectors are not capped at 1.
- Well-known ones:  $V_{\text{derived}}$  (the sum of the vectors),  $V_{\text{timestamp}}$ ,  $V_{\text{expiry}}$  (0 meaning no expiry),  $V_{\text{lastmodify}}$ ,  $V_{\text{location}}$ , and  $V_{\text{link}}$  (forward, backward, bidirectional).
- They give information atoms kinds that help group them for specific tasks; the applications named are typing and sorting.

- They can build data structures in Assembly (sequences, graphs, layered clusters) and in Inference.
- They are part of information atoms. Some change with content and some do not.

Review of that specification against the paper’s own text, with the demonstrations run. Nothing in the paper changed.

#### **Conflicts with text already in the paper.**

- *G1, label identity.*  $\uplus$ ,  $D$  and  $\text{UnCollapse}$  compare content. If a timestamp joins an atom’s identity, every atom is unique by birth time: three contributions  $A$ ,  $B$ ,  $A$  give  $D = 2$  by content and  $D = 3$  with timestamps, so  $N - D$  is always 0 and  $A \uplus A \neq A$ . Equality must be content-only, and a rule is needed for which labels survive when  $\uplus$  merges two atoms of equal content.
- *G2, the level rule.* §3.1 says “a unit carries no representation of what it constitutes. A seated word does not know the poem,” and that which units were co-present, over what interval, and within what region are “constituting facts of a level, not coordinates of an atom,” which is “why the atom stays at four coordinates and one sort tag.” Time, place and links sit against this. A candidate resolution: labels ride beside the four coordinates, as  $V_{\text{derived}}$  already does, and a link may point to sources or peers but not up to what the atom constitutes, since a word’s forward link to its poem would break the rule. The sentence needs rewording either way.
- *G3, the derived-vector limit.* Axiom 7 says “at most one further, derived vector may sit alongside them.” A family of labels contradicts it. Reword to at most one derived-from-content vector (one per sort once R36 lands), with recorded labels separate, all excluded from  $N$  and  $Q$ , and the seven metrics and  $V_{\text{derived}}$  ranging over informational vectors only.  $V_{\text{derived}} = \sum r_i \hat{n}_i$  with  $\chi$  excluded, which “the sum of the vectors” abbreviates.
- *G4, mutation.* The paper has no in-place change: its verbs act “alongside, not in place of,” and  $\text{Collapse}$  produces an additional atom.  $V_{\text{lastmodify}}$  and forward or bidirectional links assume an atom that persists through change, which is the identity handle (R18, not adopted). With immutable atoms only backward links are free, since a newer atom can point to older sources, which is the provenance reasoning the  $\text{Collapse}$  section already makes;  $V_{\text{lastmodify}}$  then equals the newer atom’s  $V_{\text{timestamp}}$ . One world must be chosen.
- *G5, vocabulary.* “Typed” is in the paper’s title, and type/typed occur 21 times and sort 88 times. Meta vectors giving atoms “types,” and “sorting” them, collide with  $\tau$ . Use another word for the kind (label or class) and “order” or “rank” for the verb.

#### **Gotchas in the logic.**

- *G6, timestamp encoding.* v17’s clock-face vector gives the identical vector for now, one day ago, and one week ago (checked), so age and expiry cannot be read from it. They need an absolute, non-cyclic time. This answers the earlier “one component or two”: an absolute scalar for age and expiry, and optionally the clock-face vector for grouping by time of day.
- *G7, sentinels.* “Expiry 0 means none” reads as already expired under  $\text{now} < \text{expiry}$  (checked), and a location of  $(0, 0)$  is a real place. The paper’s own idiom is that black is an empty seat, so “no expiry” is the absence of  $V_{\text{expiry}}$ , and absence is not zero.

- *G8, granularity and propagation.* Coach may mean atom or core. Timestamp, expiry and link are per core (the R38 registers), while  $V_{\text{derived}}$  is per atom, and per sort once R36 lands. A Collapse result is a new atom whose own birth differs from its cores' births, so there are two time notions. Each verb also needs a stated rule for the labels of its result; the proposal is that a result is new with its own birth, core labels travel with their cores under Collapse, and links record provenance.
- *G9, grouping needs a comparison per kind.* Sphere distance for location, cyclic distance for the clock, equality for identifiers, graph distance for links: no common metric, so each kind needs a stated rule, the analogue of Axiom 0 for labels. Links are identifiers and not coordinates, and calling labels vectors invites arithmetic; no verb acts on a label, as none acts on the seat  $\vec{P}$ .
- *G10, two sources of structure.* Links declare structure, while Assembly's energy produces camps from geometry and a seat; keep them distinct. Link use belongs to Paper 2 (R16) and the taxonomy to Paper 1. Links held in labels can also carry order past  $\uplus$ , so state that  $\uplus$  does not read labels, which keeps "order comes from outside" true of what  $\uplus$  reads.

*Status: R15 specified; R45 open.*

## v24 — rulings on the meta-vector questions

Rulings given:

- Drop  $V_{\text{lastmodify}}$ . Atoms are immutable and change only through Collapse and UnCollapse.
- Meta vectors are not part of  $D$ .
- Meta vectors link to other things mostly by criteria on other meta vectors and never by content. Words are never linked to a poem: the words do not know the poem, the path knows it.
- The absence of  $V_{\text{expiry}}$  means no expiry.
- Coach means atom, very loosely. Information atoms and meta vectors share the same cores, a core being a single point whose weight varies with the  $N$  count;  $V_{\text{derived}}$  has a magnitude and a direction that help in higher dimensions.
- Meta vectors help guide information atoms in higher dimensions, a useful attribute in Assembly and Inference.

### What each ruling settles, checked against the paper.

- *G1, label identity.* Equality is content-only: two cores coincide when they hold the same vectors, and labels do not enter  $D$ , so they do not enter  $\uplus$ 's equality or UnCollapse's value-exactness either. What  $\uplus$  does with the labels of merged atoms remains (G14).
- *G2, the level rule.* It stands unchanged. Links run from label to label, and a link never carries membership in a larger whole, which is the paper's own rule that a seated word does not know the poem.

- *G4, mutation.* Immutable atoms make backward links the free ones, and  $V_{\text{lastmodify}}$  is dropped. One wording consequence: the paper says UnCollapse “removes one core from a Collapsed atom,” which reads as mutation. It returns a new atom without that core and leaves the original unchanged, a one-line fix at import.
- *G7, sentinels.* Absence is not zero, and the rule extends to every label. Validity is then: valid exactly when  $V_{\text{expiry}}$  is absent or now  $< V_{\text{expiry}}$ .
- *G8, granularity.* Shared cores make labels per core, matching the R38 registers.  $V_{\text{derived}}$  exists per core and per atom, and the atom’s is exactly the sum of its cores’, since  $\sum_i r_i \hat{n}_i$  regroups by core.

### New points the rulings raise.

- *G11, a core as a point with weight.* A multi-vector core spans several directions, so the single point is its  $V_{\text{derived}}$ , not its content. The weight at a point is the multiplicity  $m(c)$ , the number of equal cores there, with  $N = \sum_c m(c)$ ,  $D$  the number of distinct cores, and  $N - D = \sum_c (m(c) - 1)$ . Call it multiplicity, not density, which R12 retired, and keep it a property of the atom’s list and not of the core: a core does not know its siblings.
- *G12,  $V_{\text{derived}}$  as a guide.* The paper already says operators needing one pair read the derived sum as a weight or a direction, and that beyond three dimensions it cannot be  $T$ ’s actor, which needs  $N_{\text{dim}} - 2$  orthonormal arrows. So guidance means read-only roles: a representative weight or direction, a seat, grouping. Its reliability is atom coherence,  $|V_{\text{derived}}|/S_{\text{atom}}$ , which is 0 for an equal antipodal pair although  $S > 0$ , leaving the direction undefined. The verified third candidate of v1, disjoint dimensions, which commute exactly and keep content recoverable, is the available mechanism for labels to occupy their own dimensions without disturbing content.
- *G13, links.* Links based on label criteria are derived at read time. Only backward provenance, identity-based since the identifier is itself a label (R18), is stored at birth. Forward and bidirectional are views from a reverse index, and the three  $V_{\text{link}}$  names become view directions.
- *G14,  $\uplus$  and labels.*  $\uplus$  discards order and count, so it naturally discards per-copy labels too, while Collapse keeps every copy with its labels, which is the per-core register. This is the proposed default and is unconfirmed. It also resolves R19 by proposal: filtering by expiry is UnCollapse of the expired cores, a selection by label criterion, so Mix and Collapse need not know about expiry.

The paper already holds two meta vectors without the name:  $V_{\text{derived}}$  and the seat  $\vec{P}$ , which is uncapped, sits outside the four-tuple, and enters  $d_{AB} = |\vec{P}_B - \vec{P}_A|$ . Importing the taxonomy therefore mostly names what is there, and  $V_{\text{location}}$  may turn out to be the seat.

Still open before import: the wording of the derived-vector limit (G3), the word for the kind (G5), the timestamp as an absolute scalar with an optional clock-face vector (G6, unconfirmed), the comparison rule per kind (G9), and the two sources of structure (G10). *Status: G1, G2, G4, G7, G8 settled; G11–G14 proposed.*

### v25 — confirmations, a check of the assembly claim, and bowls keeping labels

- *G11.* Confirmed: the word is multiplicity, used loosely.

- *G13*. The first part is confirmed: links by label criteria are derived at read time. The second part is accepted on this explanation: an immutable older atom cannot be edited to record a later one, so it never stores a forward pointer. “What was  $A$  used in?” is answered by searching for atoms whose backward link names  $A$ , and that search result is the forward view.
- *G12, refined and checked*.  $V_{\text{derived}}$  is the sum of an atom’s vectors, with a magnitude and a direction that assist it in higher dimensions, and the stated expectation is that atoms mostly assemble with others of similar magnitude and direction. Checked against the assembly energy  $U_{AB} = -\frac{r_{AB}}{2}(3 + \hat{n}_A \cdot \hat{n}_B)/(d_{AB}^2 + \varepsilon_{AB}^2)^{3/2}$ ,  $\varepsilon_{AB} = r_A + r_B$ :
  - *Direction: supported*. Aligned pairs pull exactly twice as strongly as opposed ones at every separation (ratio 2.000 at  $d = 0, 0.5, 1, 3$ ), and in a random population of 300 atoms the best partner averages  $12^\circ$  to  $19^\circ$  away against about  $90^\circ$  for a random partner.
  - *Magnitude: conditional*. For an atom of  $r = 0.60$  the best partner has  $r = 0.30$  at contact ( $d = 0$ ),  $0.42$  at  $d = 0.5$ ,  $0.69$  at  $d = 1$ , and  $1.00$  at  $d = 3$ . Only at fixed total strength do balanced pairs win, with  $|U|$  at  $d = 1$  of  $0.1768$  for  $(0.5, 0.5)$ ,  $0.1485$  for  $(0.7, 0.3)$  and  $0.0636$  for  $(0.9, 0.1)$ . In the population the best partner’s strength gap is smaller than a random partner’s at  $d = 0$  and  $d = 1$  ( $0.27$  and  $0.24$  against  $0.32$ ) and larger at  $d = 5$  ( $0.42$  against  $0.31$ ).

So direction is a clean grouping rule, while magnitude is a weight whose effect depends on separation relative to the units’ own extents. Two cautions:  $|V_{\text{derived}}|$  is uncapped and grows with  $Q$  and with multiplicity, so a many-vector atom is heavy in the product and in  $\varepsilon_{AB}$ ; and the direction is undefined at cancellation, where atom coherence is 0. This is a candidate for R22: read an atom’s single  $(r, \hat{n})$  from  $V_{\text{derived}}$ , which the paper already allows for operators that need one pair.

- *G14, revised*. A bowl may keep labels as well, since labels only identify the nature of atoms and  $\uplus$  has no reason to read them; they stay available to anything that wishes to. Consequence:  $\uplus$  still discards count in what it reads, and content-level quantities see one seat, but if per-copy labels ride along, count survives in them whenever the labels differ. The pack and unpack map between a bowl and a Collapsed atom (R39) is then lossless exactly when per-copy labels are distinct, for instance identifiers. Bowl and Collapse stay nearly identical but not the same: repeats enter a Collapsed atom’s content, so  $S$ ,  $\mathcal{E}$  and  $V_{\text{derived}}$  see them, and they do not enter a bowl’s. Proposed default, unconfirmed: labels carried on a seat of equal content merge as a set, equal values coinciding and distinct ones both kept.
- *G5, G6, defaults proposed*. The paper already uses “label” in the sense meant (“direction is a label here,” “a label on the relation”), so the individual meta vector is a label and atoms are grouped by their labels, with no new noun and no “type”; “ordering” replaces “sorting.” If a noun is wanted, “category” has zero uses in the paper, against “class” with four and “kind” with ten. The timestamp is an absolute scalar for age and expiry, with the clock-face vector optional for grouping by time of day.

*Status: G11, G13 confirmed; G12 refined; G14 revised; G5, G6 defaults proposed; import awaiting go-ahead.*

## v26 — meta vectors imported into the paper

Merged into `Pattern_Arithmetic_v15.25.tex` as Section 2.11, after Axiom 7. The term is “meta vector,” which had no prior use in the paper, where “meta” occurred only in “meta-rule” and in the metadata on a walk’s steps. “Label” stays a plain word: it has seven loose uses, which reserving it would have forced rewording.

**What went in.** A meta vector is not capped, not informational, counted in neither  $N$  nor  $Q$ , outside the seven metrics,  $D$  and every verb, with no  $\tau$ -justified admissibility of its own and no place among the coordinates; it rides with the cores it describes. Two were already in the paper unnamed:  $V_{\text{derived}}$ , per core and per atom with the atom’s the sum of its cores’, and the seat  $\vec{P}$ . The recorded ones:  $V_{\text{timestamp}}$  as an absolute time with age derived and a clock-face vector optional (the hour on  $\theta$  fails, as v17 showed);  $V_{\text{expiry}}$ , whose absence means none and which is valid exactly when absent or now  $< V_{\text{expiry}}$ ;  $V_{\text{location}}$ , a point on a sphere and not a sort, by the cap test and the seat precedent; and  $V_{\text{link}}$ , naming targets by another atom’s meta vector and never by content, stored backward only, with forward and bidirectional as views. Absence is not zero. Atoms are not edited, so there is no last-modified vector. Equality ignores meta vectors ( $A, B, A$  gives  $D = 2$  by content and 3 with timestamps). They group atoms read-only, with a comparison per kind; direction is the cleaner assembly rule and magnitude a weight; links declare structure while the energy produces it from geometry; and order carried in meta vectors passes  $\uplus$ , which does not read them.

**Supporting edits.** Axiom 7’s derived-vector limit reworded; the level rule of §3.1 notes that a meta vector records a fact about the atom itself and never what it belongs to; the Assembly paragraph names the seat as a meta vector; Collapse says each core keeps its meta vectors; the counts and  $D$  exclude them; UnCollapse now “returns a Collapsed atom without one core, leaving the original unchanged” where it said “removes one core from”; the identity handle is named as a meta vector; and the open item on location as a sort is removed, since the subsection settles it.

**A correction found along the way.** R22 had been recorded as “Axiom 6 silent,” but Axiom 7 already says that  $V_{\text{derived}}$ ’s magnitude and direction stand in for the atom wherever a weight or a direction is required, naming Assembly’s  $r_{AB}$  and a camp body’s alignment. R22 is closed in substance, and v25’s “candidate for R22” described what the paper already says. The cost that  $|V_{\text{derived}}|$  is uncapped, and so heavy in the product and in the softening length, follows from that existing sentence and not from this import.

**Checks.** All 49 numbered equations are identical to v15.24’s; zero errors and undefined references; the same five overfull warnings; and the seven metrics and  $Q_{\text{eff}}$  occur nowhere outside Axiom 7, the new subsection using none of them.

**Not merged.** G11 (multiplicity as a name), the merge-as-set default of G14, R19’s filtering by UnCollapse, a per-sort  $V_{\text{derived}}$  (R36), and the question of an atom-level birth time distinct from its cores’ births. *Status: R14, R15, R22, R45 closed; R16, R19 partly merged.*

## v27 — rulings on form, expiry, Collapse and identity; fixes 1 to 3 merged

Rulings given:

- Meta vectors are always vectors. Where one must act as a scalar it may exist in  $\mathbb{R}^2$ , and standard planes (horizontal and vertical) are wanted as recommendations for the various scalar vectors, with a registry for scalars as the idea.
- $V_{\text{derived}}$  always rests on information-content strength. If an atom has an expiry, all its meta vectors become meaningless and it no longer takes part in any process. If an inner atom of a larger Collapsed atom expires, all its information and meta vectors stop participating.

- A larger Collapsed atom collects all its inner atoms’ information vectors at standard position and their meta vectors.  $V_{\text{derived}}$  is recalculated over the denser content. The result has a new timestamp vector and also holds the timestamps linked to each core. Neither information vectors nor meta vectors ever lose identity, under Collapse, Bundle or Mix.
- Meta vectors are per core and, like information atoms, do not lose identity. If an atom expires, all its information and meta vectors stop contributing.

**Merged in v15\_26, the three fixes asked for.** The “at most one” derived vector in Axiom 7 now reads one per core and one for the atom, the atom’s being the sum of its cores’; the section’s “part of identity” now says “part of the value,” and separates value from the identity handle, which tells sources apart; and granularity is stated for location and link as well, with a paragraph on where meta vectors are held, including a Collapse result’s own new timestamp and backward link beside its cores’. That last paragraph also answers the open question of an atom’s birth against its cores’ births.

**What the rulings raise, checked.**

- *R46, planes and a registry.* A registry fits as a further catalogue in the paper’s deferred-catalogues subsection, which already holds the concept dictionary and the lexicon as recorded and left open, not an axiom; it says “a pair of catalogues” but lists three items, the third not a catalogue. Recommendations only, as standardization of the lexicon is. On geometry: two planes inside a 3-D stage always share a line (checked on 20,000 random pairs), so “horizontal and vertical” planes cannot be disjoint there. Disjoint planes need extra dimensions ( $N_{\text{dim}} \geq 3 + 2k$  for  $k$  planes; coordinate planes  $(x_1, x_2)$  and  $(x_3, x_4)$  in  $\mathbb{R}^4$  meet only at the origin), which is the mechanism of v1’s third candidate, shifts commuting exactly with content. Or each plane is a separate space attached to the core, needing no ambient dimensions. Within a plane, a scalar on one axis keeps vector distance equal to the difference (0.500 for values 0.5 apart). A polar form, radius as absolute time and angle as time of day, packs both components into one vector but breaks that: for times 24, 12 and 0.5 hours apart the distances are 24.000, 2012.000 and 130.840, so age must be read from the radial part. Suggested convention: the scalar on the plane’s first axis, a cyclic companion as the angle if wanted, and planes named by role (a time plane, a place plane), with horizontal and vertical a drawing convention only.
- *R47, expiry as read-time exclusion.* Excluding expired cores from  $V_{\text{derived}}$ , the metrics and every process makes derived quantities functions of content and of time. The paper says every law is deterministic, the same starting units giving the same result, so processes need an explicit time argument  $t$ , defaulting to now. The counts need a rule: stored  $N$  and  $Q$  against active ones. A proposal that reuses what exists: an expired core reads as black, counted in  $N$  and absent from  $Q$ ,  $S$  and  $V_{\text{derived}}$ , which is the silent-contribution rule. An atom-level expiry silences all its cores; an inner atom’s expiry must reach its cores after Collapse keeps them distinct, so either it is copied to each core at Collapse or the grouping is kept, the minimal reading being to copy it down at birth. A link to an expired atom is lineage and not participation, and the proposal is that provenance stays readable. The atom itself never changes: exclusion is a read.
- *R48, two levels.* A Collapse result carries its own new timestamp and its cores’ own, so meta vectors exist at both levels. Open: whether the atom level also has expiry and location, and how validity nests.
- *R49, “never lose identity.”* The word now has three senses: value (equal content), individual (this thing with its own meta vectors), and handle (an identifier). The ruling means that

individuals persist. Collapse keeps cores inside the result. Bundle keeps individuals as meta vectors at the seat: value-equal ones share a seat, count is discarded as content and survives in the labels when they differ. Mix leaves its sources unconsumed, since it is a reading, while the result fuses their values; UnMix stays impossible, as the paper proves the sources are not determined by the result. So “never lose identity” cannot mean recoverable from the result. It means persistence of the sources, named by the result’s backward links, and the paper should say so to avoid appearing to contradict its own UnMix proof.

*Status: fixes 1–3 merged; R46, R47, R49 open; R48 partly merged.*

## v28 — expiry, identity and the scalar convention merged

Merged into `Pattern_Arithmetic_v15_27.tex`, on the rulings that expiry follows the proposal, that UnMix stays impossible, and that scalars live on designated axes with no other constraint on meta vectors.

### What went in.

- *Form (R46)*. A meta vector is a vector with no constraint on its form. A scalar one takes a designated axis of its own space, uncapped, so that vector distance is the difference of values, and uses a plane or more where a kind needs more information. It lives in its own space or in axes disjoint from the content’s, so that no verb that rotates content touches it; that is a requirement of “no verb acts on a meta vector,” not a constraint on shape. Which axis suits which kind is a recommendation: a meta-vector catalogue beside the dictionaries in the deferred-catalogues subsection, which now says “a set of catalogues, with learnable paths beside them” where it said “a pair.”
- *Expiry (R47)*. Everything is read at a time  $t$ , by default now, and the determinism sentence now says so. A core is valid at  $t$  when  $t$  is earlier than every expiry it carries. An atom’s own expiry applies to every core in it, and a contribution’s expiry is carried onto each of its cores at Collapse, beside their own. An expired core contributes no vectors and reads as a black core.  $N$  and  $D$  are counts of the stored atom and do not change with  $t$ ;  $Q$ ,  $V_{\text{derived}}$ , the metrics and every process read only the valid cores. Expiry is a reading, not an edit, and backward links stay readable.
- *Identity (R49)*. The individual persists under Collapse, Bundle and Mix, as persistence and not recovery; UnMix stays impossible.

**Two wording choices to confirm.** First,  $D$  is a stored count and does not follow  $t$ : were it to, all expired cores would coincide as one black value and  $D$  would shrink as cores expire. Second, under Bundle the information content of value-equal individuals shares a seat by definition, so what persists of the individuals is their meta vectors; the ruling that information vectors never lose identity can only mean this, since  $A \uplus A = A$ .

**A consequence noted.** A core can now carry several expiries, its own and inherited ones, so the bullet’s “ $V_{\text{expiry}}$ , per core” reads as each expiry the core carries.

**Checks.** All 49 numbered equations are identical to v15\_26’s; zero errors and undefined references; the same five overfull warnings; and the seven metrics are absent outside Axiom 7, the meta-vector subsection using none of them. *Status: R47, R49 closed; R46 and R48 partly merged.*

## v29 — a rigid turn carries the meta vectors; two corrections to v15\_27

Ruling: a meta vector's axis shares the atom's cores. Turn an atom and all of its vectors, information and meta, turn with it,  $V_{\text{derived}}$  included. Meta vectors are labels on the coach and move with it; relative positions and distances are kept, so the meaning does not change.

**Checked.** Turning an atom as a whole, the same rotation on every direction and the same phase shift on every phase, gives a  $V_{\text{derived}}$  equal to the turned  $V_{\text{derived}}$ .  $|V_{\text{derived}}|$ ,  $S$ ,  $\mathcal{E}$ ,  $\alpha$ , every pairwise angle and every phase difference are unchanged, and only the direction of  $V_{\text{derived}}$  changes. For a scalar placed on an axis of the same ambient space as the content, rotating the whole space preserves distances but changes the value, a timestamp coordinate going from 1.7600 to  $-0.6797$ ; rotating only the content axes leaves it at 1.7600. So the reading adopted is that a scalar meta vector is a coordinate in the atom's own frame, carried unchanged, while  $V_{\text{derived}}$  turns with the content.

**Two corrections to my v15\_27 text.** First, it said a meta vector lives in its own space or in axes disjoint from the content's, so that no rotation touches it. That is wrong for  $V_{\text{derived}}$ , which lives in the stage and turns with the content. Second, it said every verb leaves a meta vector alone, but Assembly reads  $V_{\text{derived}}$  and the seat. The text now says no verb changes one or takes one as an operand, though a verb may read one.

**A caution carried into the wording.** Distances between a content vector and a scalar axis mix units, a unit direction against a time, so preservation is claimed only among the information vectors and  $V_{\text{derived}}$ .

**Not yet defined elsewhere in the paper.** Turning a whole atom.  $T$  acts on one occupant, so the text defines the turn as the same rotation and phase shift applied to each vector. It sits beside R21 and R40.

**Checks.** All 49 numbered equations are identical to v15\_27's; zero errors and undefined references; the same five overfull warnings; and the seven metrics are absent outside Axiom 7 and from the subsection. *Status: R46's form corrected.*

## v30 — meta vectors need to be consistent, not meaningful

Principle given: meta vectors are properties put to use mainly in Inference and Assembly. They need only be consistent and mathematically accurate. They need not make sense by themselves, as a word need not make sense by existing; their use is extracted by context and by the path, poem or prose.

**How it sits with the paper.** It is the paper's own stance applied to labels: a seated word does not know the poem, and the algebra "does not mint the relations." It also explains why meta vectors need no Axiom 0 admissibility, since nothing in the algebra reads them for meaning.

**What it changes in my earlier assessment.** I counted the lack of a worked example for timestamp, expiry, location and link as a maturity gap. Under this principle the test is consistency and accuracy, and the uses belong to Assembly and Inference, so the remaining obligations concern consistency only.

**What it does not remove.** "Need not make sense" does not mean "may be underdetermined." A comparison across mixed units depends on the unit chosen. Take  $A$  at  $0^\circ$  with time 0,  $B$  at  $10^\circ$  with time 3000s, and  $C$  at  $60^\circ$  with time 100s. The nearest neighbour of  $A$  is  $C$  with time in seconds (100.0 against 3000.0) and  $B$  with time in hours (0.851 against 1.000) or in days. So any use that compares across kinds must state its comparison rule and unit. That is the work of the catalogue (R46) and of the comparison per kind: the vector need not make sense, but its convention must be fixed.

### Remaining consistency obligations (R50).

- The labels of a result of  $T$ , offset, Remove and Assembly, which the paper states only for Collapse, Bundle and Mix. In particular, whether a turned atom, being a new atom, gets its own new timestamp and backward link as a Collapse result does, while its cores keep theirs.
- Conventions that fix values: the time standard (UTC, resolution) and the axis and unit of each scalar kind.
- $V_{\text{derived}}$  per sort once mixed sorts are admitted (R36).

**Proposed, not merged.** One sentence for Section 2.11: “A meta vector need not mean anything by itself. It must be well defined, consistent, and mathematically accurate; what it is good for is extracted by the context or the path that reads it, as a word’s use is.” If R28 splits the paper, the “What they are for” paragraph moves with Assembly and Inference. *Status: principle recorded; R50 open.*

### v31 — the Mona Lisa again, and colour as a position

Question: can an information atom hold an accurate, complete representation of the Mona Lisa, as a Collapsed atom of several pure types (colours, brush strokes, hues), each backed by a standard, the standards being recommendations and not constraints?

**Answer, with a stand-in tested** (a  $32 \times 32$  image and a stroke-direction field, not the painting).

- *Capacity.* No limit on  $N$  or  $Q$  was found in the paper, and cores may be any size.
- *Fidelity.* Lossless relative to a standard and a precision. One core per canvas position ( $N = 1024$ ) round-tripped: the 8-bit colours exactly, the stroke directions to  $1.1 \times 10^{-16}$  rad, and a black pixel as  $r = 0$ , which is Axiom 2’s empty seat. “Complete” can only mean complete relative to the standards, since no finite standard exhausts a physical painting.
- *Sense.* Not stored: a path, or a lexicon pin, reads it.
- *How it is held.* A core is a point, so one core per position, each carrying its position as a meta vector. The Form paragraph allows a plane, but  $V_{\text{location}}$  as defined is geographic, so a canvas position is another kind in the catalogue. Today the paper supports one atom per sort in a bowl, aligned by position labels, which is the link rule: criteria on other meta vectors, not content. Pairing by label was 1024/1024 correct and pairing by order alone 0/1024. One atom of several sorts needs mixed-sort cores (R36), and the Mona Lisa is the test case for adopting it.
- *Standards.* A sort’s standard attaches through the catalogue. If one sort has several standards, sRGB or P3, someone must say which, and decoding needs it; it might be recorded as a meta vector.
- *Cautions.*  $V_{\text{derived}}$  is a summary and not a representation, and the pixels must never be Mixed, since Mix fuses values and UnMix is impossible whereas Collapse keeps every core.

**Colour as a position (ruling).**  $V_{\text{derived}}$  of colour positions an atom in a higher space, so summaries of like colours or shades fall in the same region. Checked:

- It works only if colour is carried by direction.  $V_{\text{derived}} = \sum r_i \hat{n}_i$  excludes  $\chi$ , so in the paper’s phase-colour convention, red ( $\chi = 0^\circ$ ) and green ( $\chi = 120^\circ$ ) at one direction have identical  $V_{\text{derived}}$ . The colour standard must put colour in  $\hat{n}$ , while phase colour stays for Mix arithmetic, so the paper has two colour roles to keep apart.
- With colour carried by direction (RGB direction, strength as brightness), dark and light red share a direction ( $0.00^\circ$ ), a near red is  $1.31^\circ$  away, and green is  $71.02^\circ$  away.
- It is many-to-one. Fifty red with fifty green, and one hundred uniform yellow, share the direction  $(0.707, 0.707, 0)$ , with atom coherence  $|V|/S$  of 0.707 and 1.000. The locator is therefore direction plus coherence.
- $|V_{\text{derived}}|$  scales with the number of cores (0.519, 51.877, 5187.748 for 1, 100, 10000 identical colours), so it measures size and not colour. Position by direction and normalized magnitudes. This also feeds the earlier caution that an uncapped  $|V|$  is heavy in the Assembly energy.

*Status: R51, R52 open.*

## v32 — review of v15\_28: simplicity, consistency, repeatability, reproducibility

Method: v15\_28 was compiled and audited, then 39 numerical claims were recomputed from the inputs the paper prints, with a script delivered as `reproduce_v15_28.py`, the supplement R29 asked for.

**Reproducibility: 36 pass, 0 fail, 3 noted.** Passing: the Axiom 7 identity and examples, the Mix example, Sun–Wind–Sky, the four-dimensional actor, the unmixing pair, the storm costume (both paths, endpoints and angles), the offset equator, the beyond-3-D order gaps (0.683 to 0.331), and the full enclosure replay: steps to freeze 628, 1261, 2526, 6320 exactly, the bodies, the four runs agreeing to  $1.97 \times 10^{-7}$ , the  $r$ -weighted body  $2.4 \times 10^{-8}$  from the plain one, and the pair falling from 3 to 1.80. The three exceptions:

- $\sigma_{\text{atom}}$  clumped 0.00032 against uniform 0.00214: the populations are not printed. A uniform spread over  $(0, 1]$  gives 0.0032, and the printed 0.00214 is close to a uniform spread over  $(0.2, 1]$  (0.0021). The populations should be printed.
- The 804 of 3000 steps on which the separation increases is rounding-sensitive: 830 to 855 for algebraically equal orderings of the same arithmetic or a  $10^{-12}$  nudge to the start. The qualitative claim is robust, hundreds of increases against none, and dividing by  $r$  instead of  $r^3$  gives none. It should be stated as a range or as more than a quarter of the steps.
- $[0.3/0.1]$  is 2 in floating point, so shell edges are ambiguous; the printed populations are unaffected. Bins should be defined in exact arithmetic or with a tolerance.

Claims of “checked directly” without inputs: the naive alternative that borrows the axis from  $x$  is not defined, and neither are the “uniform case” of shell counts or the values of  $K$  tested.

### Consistency findings.

- “Not a sixth verb” (the introduction) is stale: the paper has six.
- Table 1’s UnCollapse row and caption say “contribution” where §3.4 returns an atom without one core.
- The title banner of equals signs overflows the margin (979 pt overfull) and is clipped at the page edge.

- Axiom 7 carries a drafting-history sentence (“Earlier drafts called ... mass, density, atom energy”); the reasoning can stay as a plain statement.
- The enclosure’s “mass-weighting” and “mass-like quantity” reintroduce the word Axiom 7 retires for the  $r^3$  quantity.
- 30 of 97 labels are never referenced, which is cosmetic.
- R26 is still open: the Mix-refusal closure says “already demonstrated ... correct behavior.”
- The bowl:  $V_{\text{derived}}$ , the metrics, and expiry are defined for atoms only, and a bowl may carry meta vectors in one sentence. The quantities depend only on the multiset of vectors, so one statement over a holder would serve both. With no repeats ( $D = N$ ) an atom and a bowl coincide; with repeats they differ by multiplicity, for example  $A \oplus B \oplus A$  with  $A = \{0.8, 0.4\}$  and  $B = \{0.6\}$  has  $S = 3.0$ ,  $\mathcal{E} = 1.96$ , while the bowl  $\{A, B\}$  has  $S = 1.8$ ,  $\mathcal{E} = 1.16$ .

**Simplicity.** About 19,600 words and 48 pages, up from 45 at v15\_21. Axiom 7 (about 1,650 words) and Section 2.11 (about 1,370) together are about 15% of the text, and nothing in the paper reads the seven metrics or the recorded meta vectors; the costumes are about 4% and the enclosure about 8%. The spine, the verb table and the three equations, is intact. Candidate trims, tied to R27 and R28: move the metric catalogue,  $Q_{\text{eff}}$  and the empty-atom conventions out; keep the definition and rules of meta vectors and move the detail of the recorded kinds to Inference and Assembly; remove the drafting-history sentence. *Status: supplement delivered; R53–R55 open.*

## 4 Merged directly, without an ideas entry

These went straight into the paper at your direction and are already in `Pattern.Arithmetic.v15_22.tex`: the Pāṇini expansion (pratyāhāra and vipratīṣedha as structural parallels, with the Ingerman 1967 attribution of “Pāṇini–Backus form” and the oversimplification caveat); the paragraph on summing separate realities; the quantum-mechanics/general-relativity paragraph, deliberately hedged so that it does not claim Pattern Arithmetic bridges them; and the  $N/D$  addition (filed afterwards as R5).

## 5 What remains, in dependency order

1. **Done in v15\_25:** the meta-vector taxonomy and the v17 resolution (R15, R14, R45), which closed Gap 2. What remains of that cluster: R19’s filtering by UnCollapse, the label-merging default of G14, and populating the meta-vector catalogue (R46).
2. **Decide the scope questions (R27, R28):** Axiom 7’s metrics and the costumes, and whether to split. These also decide whether R12, R25, and R41–R43 still matter for Paper 1.
3. **The atom’s structure (R20–R22, R36–R40).** The word is settled (R38), and cores of any size, the empty atom, and Collapse keeping cores distinct are merged (R37, R44; v15\_24). What remains is mixed sorts, per-vector  $\tau_i$  (R36), which closes R20 in the same step. R21 can be closed by a sentence, whether a stated refusal or a stated rule (R22 is closed in substance), and the commuting test (R39, R40) can then be run.
4. **Package the Inference-bound items (R17, R30–R35)** as one coherent addition to the Inference paper: the candlestick, the technique-selection argument, the spherical Douglas–Peucker adaptation, the sphere-packing distance, and meta-vector grouping.

5. **Small residuals:** the one-sentence R26 clause, the review's presentational items (R29), shell-indexed energy (R23), the antipodal special case (R24), and whether  $S_{\text{atom}}/N$  earns a name.